Practice Quiz 10C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 10

Proving bounds for a recursively defined sequence.

1. Let $a_1 = a_2 = 1$, and define

$$a_n = \frac{a_{n-1} + a_{n-2}}{3} \qquad (n \geq 3).$$

Prove by strong induction that $0 < a_n \leq 1$ for every integer $n \geq 1$. State the complete inductive hypothesis and check both initial cases.

Solution We use strong induction. Let $n \geq 1$ and suppose

$$0 < a_k \leq 1 \quad \text{for every integer } k \text{ with } 1 \leq k < n.$$

For $n = 1$ and $n = 2$, the corresponding term is 1, so the claimed bounds hold in both cases.

Suppose $n \geq 3$. The indices $n - 1$ and $n - 2$ are at least 1 and less than n, so the hypothesis gives

$$0 < a_{n-1} \leq 1 \qquad \text{and} \qquad 0 < a_{n-2} \leq 1.$$

Adding and dividing by the positive number 3 yields

$$0 < \frac{a_{n-1} + a_{n-2}}{3} \leq \frac{2}{3} \leq 1.$$

The middle expression is a_n, so $0 < a_n \leq 1$. All cases have been proved, and strong induction establishes the bounds for every integer $n \geq 1$. ∎

Solutions revised using GPT-6 Astra in the Codex harness.